Internal exposure to pollutants and sexual maturation in Flemish adolescents.

Flanders is densely populated with much industry and intensive farming. Sexual maturation of adolescents (aged 14-15 years) was studied in relation to internal exposure to pollutants. Serum levels of pollutants and sex hormones were measured in 1679 participants selected as a random sample of the adolescents residing in the study areas. Data on sexual development were obtained from the medical school examination files. Self-assessment questionnaires provided information on health, use of medication and lifestyle factors. In boys, serum levels of hexachlorobenzene (HCB), p,p'-DDE and polychlorinated biphenyls (sum of marker PCB138, 153 and 180) were significantly and positively associated with pubertal staging (pubic hair and genital development). Higher levels of serum HCB and blood lead were associated with, respectively, a lower and a higher risk of gynecomastia. In girls, significant and negative associations were detected between blood lead and pubic hair development; higher exposure to PCBs was significantly associated with a delay in timing of menarche. Environmental exposures to pollutants at levels actually present in the Flemish population are associated with measurable effects on pubertal development. However, further understanding of toxic mode of action and sensitive windows of exposure is needed to explain the current findings.